

# Measuring Policy Uncertainty from Text: A Reference Guide

*How researchers turn language into numbers — and how you can too*

---

## Why Measure Uncertainty from Text?

Economic agents — households, businesses, investors — make decisions based not just on current conditions but on their expectations about the future. When policy is uncertain, people delay purchases, investment slows, and markets become volatile. Measuring that uncertainty quantitatively allows researchers to test these relationships empirically. The challenge is that uncertainty lives in language, not numbers. The solution is text analysis.

## The Keyword-Count Approach

The simplest and most widely used method for measuring policy uncertainty from text is keyword counting. The idea is straightforward: find documents that discuss a topic of interest (e.g., housing policy), identify sentences that contain both topic-related words and uncertainty-related words, and compute a score based on the frequency of those co-occurrences.

The formula used in this case study is:

```
Uncertainty Score = (# of uncertainty terms) / (# of sentences containing both  
housing AND uncertainty terms)
```

This produces a score between 0 and 1. A score near 0 means very little uncertainty language relative to housing discussion. A score near 1 means almost every qualifying sentence contains an uncertainty term. Scores around 0.05 are considered neutral.

## What Counts as an Uncertainty Word?

Common uncertainty-related terms include:

```
uncertain, uncertainty, unclear, unknown, unpredictable, ambiguous, volatile,  
risk, concern, doubt, question, may, might, could, pending, tentative,  
unresolved, contingent, fluctuate, instability
```

You should define your word list explicitly in your code and document it in your report. Different word lists produce different scores — this is a modeling decision you need to justify.

## What Counts as a Housing Word?

Common housing-related terms include:

```
housing, home, house, property, real estate, rent, rental, affordable, zoning,  
development, permit, construction, tenant, landlord, neighborhood, residential
```

## Limitations of This Approach

**Context-blindness** — Keyword counting cannot distinguish between 'we are uncertain about housing prices' and 'we are not uncertain about housing prices.' Negation is invisible to simple counts.

**Document quality** — Meeting minutes are summaries, not transcripts. Uncertainty expressed verbally may not appear in the written record.

**Word list subjectivity** — Different researchers choose different word lists, producing different scores. There is no single correct list.

**Small samples** — Monthly meeting minutes give you at most 12 observations per year, limiting statistical power.

## More Advanced Methods

If you want to go further, more sophisticated approaches include VADER sentiment scoring (which handles negation and emphasis), topic modeling (LDA) to identify uncertainty as a latent theme, and large language models (LLMs) that can classify sentences by meaning rather than keywords. For this case study, the keyword-count approach is appropriate and interpretable.

---

*Key Reference: Baker, S.R., Bloom, N., & Davis, S.J. (2016). Measuring Economic Policy Uncertainty. Quarterly Journal of Economics, 131(4), 1593-1636. <https://policyuncertainty.com> | Prepared for DS 4002 Case Study, Spring 2026*